

DATA FLOW DIAGRAM

Project Title

Rising Waters

Project Description

Rising Waters is an AI-powered Flood Prediction System that predicts flood risk using rainfall and weather parameters. The system collects user input, processes the data through a trained Machine Learning model, and provides an accurate flood risk prediction.

Introduction

The Data Flow Diagram (DFD) explains how data moves through the Rising Waters application. It shows the interaction between the user, the web application, the Machine Learning model, and the prediction result.

Data Flow Process

Step 1: User Input

The user enters environmental parameters such as annual rainfall, temperature, humidity, cloud cover, and seasonal rainfall into the application.

Step 2: Data Validation

The application validates all input values to ensure that the entered data is complete and within the expected range.

Step 3: Data Processing

The validated data is preprocessed and sent to the trained Machine Learning model for prediction.

Step 4: Machine Learning Prediction

The XGBoost model analyzes the input data and predicts whether the flood risk is HIGH or LOW.

Step 5: Result Generation

The application generates the prediction result and calculates the flood risk status.

Step 6: Output Display

The prediction result is displayed on the dashboard for the user to review and make decisions.

Input Data

- Annual Rainfall
- Jan–Feb Rainfall
- Mar–May Rainfall
- Jun–Sep Rainfall

- Oct–Dec Rainfall
- Temperature
- Humidity
- Cloud Cover

Processing

- Input Validation
- Data Preprocessing
- Feature Selection
- Machine Learning Prediction
- Result Generation

Output

- Flood Risk Prediction
- Alert Status
- Decision Support Information

Advantages

- Accurate flood prediction
- Fast processing
- Easy-to-use interface
- Supports disaster management
- Improves emergency response
- Reduces flood-related losses

Conclusion

The Data Flow Diagram illustrates how Rising Waters processes environmental data and generates reliable flood predictions using Machine Learning. The system enables authorities to take preventive action before flood situations become severe.